OFFICIAL OPERATIONS & USER MANUAL DOC-CMMS-REG-2026-V1

CMMS Register-Based Maintenance Management System

Complete Operational Guide: User Roles, Step-by-Step Workflows, Registers & Digital Forms

SOFTWARE RELEASE	APPLICABLE SCOPE	LOCAL ACCESS URL
Atlas CMMS Enterprise v1.8.0	Facility Maintenance & Asset Registers	http://localhost:3000/auth/login

1. Executive Overview & System Concept

The **Register-Based Maintenance Management System** is specifically engineered to digitally replace physical paper maintenance registers and clipboard slips with a unified, tamper-evident digital architecture.

Facilities traditionally maintain three disconnected paper records:

1. **The Physical Booking Logbook:** Located at the front desk or reception, where occupants report defects by hand.
2. **The Maintenance Work Register:** Kept by supervisors to track technician job allocations, time taken, and pending spare parts.
3. **The Maintenance Work Order Slip:** Carried on clipboards by technicians, hand-signed upon job completion.

This CMMS directly reproduces every column, field, and workflow of those physical registers while adding auto-incremented Booking IDs (#2100+), real-time status transitions, digital stylus/touch signatures, Excel/PDF exports, and live KPI analytics.

2. Verified System Login Credentials

Three pre-configured accounts are active in the database for immediate testing and operations:

Role	Email	Password	Operational Scope & Responsibilities
Super Admin	`superadmin@test.com`	`pls_change_me`	System Super Administrator with multi-tenant company and database oversight.
Administrator	`admin@cmms.local`	`Admin@12345678`	Company Admin with full authority: create bookings, assign technicians, approve & close work orders, export data, and monitor KPI analytics.
Technician / Worker	`worker@cmms.local`	`Worker@12345678`	Field Maintenance Worker: view assigned jobs, record start/end dates and completion time, report pending reasons, and sign work order forms.

✅ Operational Verification

Both the React frontend (`http://localhost:3000`) and the Spring Boot backend (`http://localhost:8080`) are running with PostgreSQL database connectivity. All three accounts are verified for seamless authentication.

3. User Roles & Exact Workflows (What Each User Does)

🏢 Role 1: Administrator

`admin@cmms.local`

The Administrator is the facility coordinator managing complaints, assignments, and ticket closures.

- **Logging Complaints:** Navigates to `/app/maintenance-booking-register` and clicks **"New Booking"**. Enters the reporting person, building, floor, room, and defect description.
- **Automatic Booking ID:** The system automatically issues a sequential Booking ID starting at `#2100`.
- **Dispatching Technicians:** Assigns an available technician (e.g. *Rajesh Technician*). The booking status automatically advances from `PENDING` to `ASSIGNED`.
- **Monitoring Operations:** Monitors the *Maintenance Work Register* to verify whether jobs are progressing or blocked by pending parts.
- **Supervisory Closure:** Once the technician finishes work, the Admin verifies the repair, enters **Closing Remarks**, assigns an official **Closed ID** (e.g., `CL-2100`), and marks the ticket `CLOSED`.

👷 Role 2: Maintenance Technician / Worker

`worker@cmms.local`

The Field Technician executes repairs on-site, records operational progress, logs blockers, and signs off.

- **Checking Assigned Jobs:** Opens `/app/maintenance-work-register` to see jobs assigned to their name.
- **Commencing Work:** Upon starting the repair, records the **Starting Date & Time** and updates status to `IN PROGRESS`.
- **Logging Blockers (Pending Reasons):** If parts are missing (e.g., waiting for valve or replacement compressor), the technician records the exact reason in the **Pending Reason** column so facility managers can expedite procurement.
- **Concluding Work:** Records the **Finishing Date** and total **Time Taken** (e.g. `2 hrs 15 mins`), changing status to `COMPLETED`.
- **Digital Signature:** Opens `/app/maintenance-work-order`, checks the work order row, and signs with a stylus, mouse, or touchscreen on the digital signature canvas.

👑 Role 3: Super Administrator

`superadmin@test.com`

The Super Admin governs multi-tenant instances, campus divisions, user permissions, and database health.

- Provisions facility divisions and campus accounts across multi-site organizations.
- Configures security controls, JWT expiration settings, and custom role permissions.
- Oversees system health, audit logs, and global database operations.

🔍 Role 4: Supervisor

`Assigned Supervisory Staff`

The Supervisor provides independent quality assurance and technical compliance.

- Performs physical site inspections to confirm repairs meet safety and quality standards.
- Reviews technician pending logs to approve spare parts requisitions.
- Co-signs the printable maintenance work order form.

Role 5: Staff / Complaint Reporter

`Building Occupants & Department Staff`

The Complaint Reporter notifies the facility team of issues.

- Reports defects: location (Building, Floor, Room) and description of the breakdown.
- Receives the Booking ID to track repair status through completion.

4. Module Walkthrough & Field Specifications

A. Maintenance Booking Register

Route: `/app/maintenance-booking-register`

Replicates the physical reception desk maintenance booking register:

Column Header	Type	Description & Format
S.No	Auto Counter	Sequential row reference (1, 2, 3...)
Booking ID	Unique String	Unique system identifier (e.g. `2100`, `2101`)
Date	Date Picker	Complaint registration date (e.g. `01/07/2026`)
Reported By	Text Field	Name of reporting person (e.g. *Nandhini*, *Chandramohan*)
Location Details	Compound Text	Building, Floor, Room, and exact location (e.g. *B1 – Ground Floor*)
Complaint Details	Text Area	Description of issue (e.g. *Ground floor water tap leakage*)
Booking Status	Status Badge	`PENDING`, `ASSIGNED`, `IN_PROGRESS`, `COMPLETED`, `CLOSED`

Available Actions: Search by Booking ID / Reported By, filter by Status Badge, **Export to Excel (.xlsx)**, and **Print Clean Tabular Register**.

B. Maintenance Work Register

Route: `/app/maintenance-work-register`

Records all work performed against an existing booking:

Column Header	Type	Description & Format
S.No	Auto Counter	Sequential record number
Booking ID	Lookup Dropdown	Selects booking; automatically pulls location & complaint details
Type of Work	Text / Category	Plumbing, Electrical, Carpentry, HVAC, Mechanical
Attended By	User Dropdown	Assigned maintenance technician or staff member
Starting Date	Date / Time	Date and time repair work commenced
Finishing Date	Date / Time	Date and time repair work was completed
Time Taken	Text / Duration	Duration spent on task (e.g. `1 hr 30 mins`)
Pending Reason	Text Area	Explanation if work is incomplete (e.g. *Awaiting valve replacement*)
Closed ID	Text Field	Official closure reference code (e.g. `CL-2100`)

C. Maintenance Work Order Form (Printable Paper Slip)

Route: `/app/maintenance-work-order`

A digital reproduction of the physical maintenance clipboard sheet, optimized for print and PDF export:

```
+-------------------------------------------------------------------------------------+
|                                MAINTENANCE WORK                                     |
|  Form No: 2100                        Date: 01/07/2026          Name: Nandhini     |
+------+-----------+--------------+---------------------+-----------+--------------+-------+
| S.No | Booking ID | Starting Date | Work Details        | Booking By | Finished Date | Sign  |
+------+-----------+--------------+---------------------+-----------+--------------+-------+
|  1   | 2100       | 01/07/2026    | Water tap replaced  | Nandhini   | 01/07/2026    | [✓]   |
|  2   | 2101       | 01/07/2026    | Pipe gasket fixed   | Chandra    | 01/07/2026    | [✓]   |
+------+-----------+--------------+---------------------+-----------+--------------+-------+
| Technician Signature: [_____________]       Supervisor Signature: [_____________]|
+-------------------------------------------------------------------------------------+
```

- **Multi-Row Work Table:** Technicians can record multiple completed tasks on a single printable sheet.
- **Digital Signature Pad:** Allows technicians and supervisors to sign digitally using a mouse or touchscreen.
- **Print & PDF Download:** Built-in print styles hide navigation sidebars and format the form cleanly for A4 printing.

D. Maintenance KPI Dashboard

Route: `/app/maintenance-dashboard`

High-level command center with seven real-time operational KPI cards and visual summaries:

- **Total Bookings:** Total volume of logged complaints.
- **Pending Complaints:** Unattended tickets awaiting technician allocation.
- **Assigned Work Orders:** Dispatched jobs ready for technician commencement.
- **In Progress:** Active on-site maintenance jobs.
- **Completed Work:** Finished repairs awaiting supervisor audit and closure.
- **Closed Work Orders:** Officially certified and closed tickets.
- **Overdue Maintenance:** Tickets exceeding SLA resolution timeframes.
- **Technician & Building Breakdown:** Tables displaying technician resolution speed and building defect concentrations.

5. End-to-End 11-Step Standard Operating Procedure (SOP)

1. **Complaint Reporting:** Staff notices a defect (e.g. *Ground floor water tap leakage*) and notifies facility desk.
2. **Booking Creation:** Admin creates a booking in the *Maintenance Booking Register*.
3. **Booking ID Generated:** System generates a unique Booking ID (e.g. `2100`).
4. **Technician Assignment:** Admin assigns the work to a technician. Status changes to `ASSIGNED`.
5. **Work Queue Review:** Technician logs in, views assigned work in the *Maintenance Work Register*.
6. **Work Commenced:** Technician starts work and enters the Starting Date. Status changes to `IN PROGRESS`.
7. **Progress Update:** Technician updates work details, materials, and steps taken.
8. **Pending Logging (if blocked):** If incomplete due to missing parts, technician logs the Pending Reason.
9. **Completion & Signing:** Technician logs Finishing Date, Time Taken, and signs the *Work Order Form*. Status updates to `COMPLETED`.

10. **Supervisor Verification:** Supervisor verifies completed work, enters Closing Remarks and Closed ID (e.g. `CL-2100`).
11. **Final Closure:** Ticket is marked as `CLOSED`. Booking Register, Work Register, and KPI Dashboard update automatically.

> 📌 **System Availability Note**
>
> This complete operating manual is saved as a PDF document on the Desktop and project directory for team distribution.